Politecnico di Milano

Prova finale: Introduzione all'analisi di missioni spaziali

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1. Introduction

The purpose of this work is to define some possible transfers from one point to another in space. Our team considered efficiency as a guideline, that means using a low amount of fuel, in order to minimize the weight of the vehicle, but also accomplishing the mission in an adequate time. We decided to consider also the number of impulses used; in fact, applying the velocity variation at the right time and in the selected direction requires a precise knowledge of the position and accurate attitude control; for this reason, it is desirable to have a low number of impulses to be achieved. These three requirements translate into finding the trajectories that allow us to connect the two points given using the lowest Δv possible, the fastest route and the lowest number of impulses possible. In this work we analyze different solutions, in order to find the most efficient orbital transfer.

2. Initial orbit characterization

2.1
The initial orbit is defined by position and velocity of the initial point in the ECI coordinate frames:

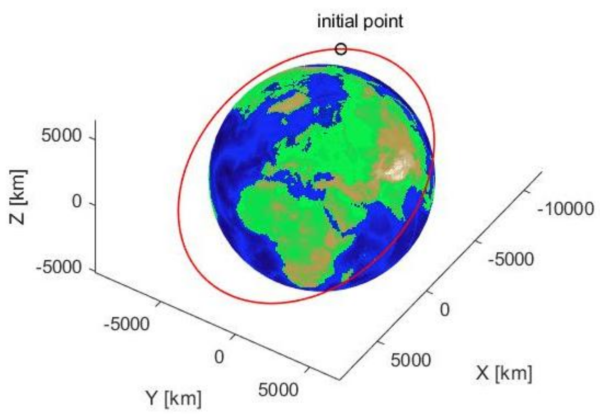
Initial point position [km]			Initial point velocity [km/s]		
x	y	z	Vx	Vy	Vz
-7021.3654	-3309.2827	2751.1479	0.9924	-6.3810	-2.4610

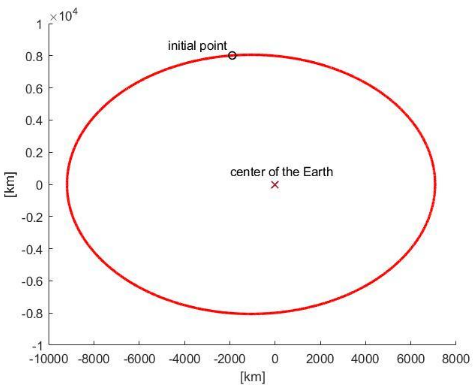
From this data we can determine the initial orbital parameters (semimajor axis **a**, eccentricity **e**, inclination **i**, right ascension of the ascending node **Ω**, argument of periapsis **ω**) and the true anomaly (**ϑ**) of the initial point:

a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	ϑ [rad]
8127.3041	0.1303	0.5507	1.0556	0.6472	1.8021

2.2
This orbit has a period of 7292 s (≃ 2 hours) and belongs to the (not firmly defined) boundary between low Earth orbit (LEO) and medium Earth orbit (MEO). The inclination, that is of approximately 31,55°, is typical of launches from the Palmachim Air Force Base located in Negev Desert, Israel (Latitude 31.5 N).

2.3
The following images are respectively the graphical representation of the orbit in space and in the orbital plane:





3. Final orbit characterization

3.1

The final point is defined by the parameters of its orbit and by its true anomaly:

a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]	ϑ [rad]
14270.0000	0.3812	1.0030	2.8600	1.9690	3.0770

From this data we can determine position and velocity of the final point in the ECI coordinate frames:

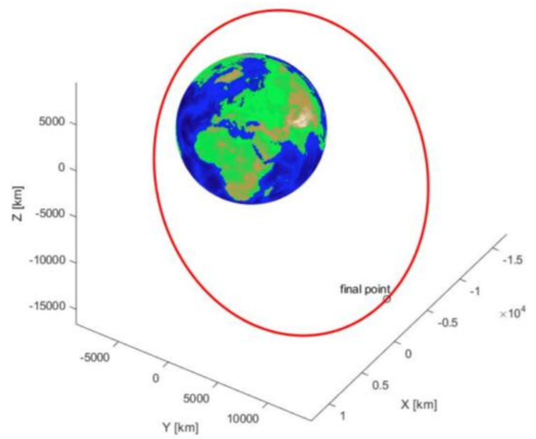
Final point position [km]			Final point velocity [km/s]		
x	y	z	Vx	Vy	Vz
-3412.4719	11399.4242	-15680.7267	-3.4127	0.4123	0.8658

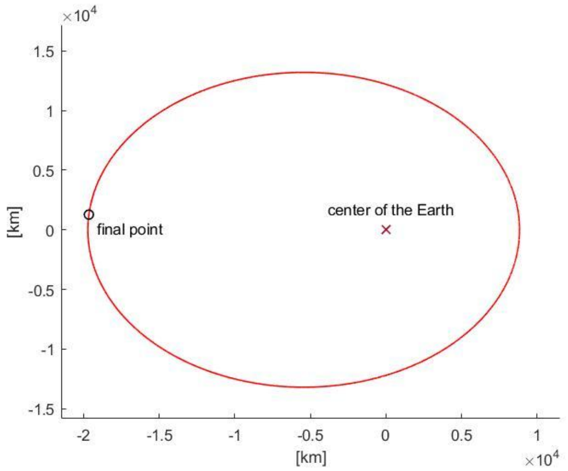
3.2

The orbit has a period of 16965 s (4h:42m:45s) and can be classified as a medium Earth orbit (MEO). The inclination, of approximately 57,46°, and the altitude, ranging from 2452 to 13332 kilometers, makes a satellite in this orbit always able to be in sight of the portion of Earth between the Artic Circle and the Antarctic Circle.
It is also worth noting that the final point has a true anomaly of 176,3° and therefore is really close to the apocenter. This fact implies that if we reach the final orbit at the apogee (which is common usage for example in elliptical transfers), we will need to wait almost an entire orbital period before reaching the final point.

3.3

The following images are respectively the graphical representation of the orbit in space and in the orbital plane:





4. Transfer trajectory definition and analysis

4.1

In this report we will discuss four different transfer strategies, one of each has the purpose of making the transfer more efficient.

1. Standard strategy
2. Optimized standard strategy
3. Transfer with circularization
4. Two-impulse transfer

4.2

1. **Standard transfer**: It consists of three different maneuvers; each one allows us to change an orbital parameter in order to reach the final orbit. The first one is the plane change maneuver; as two confocal non-coplanar orbits have two points of intersection, we characterize the plane change maneuver for both points:

	First intersection point	**Second intersection point**
ECI intersection position [km]	(7224.4; 394.6; -3741.5)	(-6952.8; -379.7; 3600.9)
Δt (initial point→intersection point) [s]	6858	3809
Δv [km/s]	8.1689	7.8619

We choose to maneuver at the second intersection point, characterized by the lowest cost and Δt from the initial point. Once the orbit's inclination is changed, we must modify the argument of periapsis. As the plane change, we can maneuver in two different points.

	First intersection point	**Second intersection point**
ECI intersection position [km]	(3525.2; -5365.8; 6545)	(-2791.9; 4143.1; -5053.6)
Δt (plane change point →intersection point) [s]	5094	1547
Δv [km/s]	0.2983	0.2983

Since this maneuver's cost does not depend on the point at which it is executed, we choose the case which minimizes the transfer time.

Finally, we reach the final orbit's shape by a two-impulse bitangent maneuver. We analyze an apogee-apogee transfer and a perigee-perigee transfer.

	First maneuver point (perigee)	**Second maneuver point (apogee)**
ECI position [km]	(-1659.2; 4127.4; -5492.9)	(2156.4; -5364.2; 7138.9)
Δt (change perigee anomaly point →maneuver point) [s]	7147	3502
Δt (maneuver) [s]	3527	8641
Δv1-first impulse [km/s]	0.0691	1.5507
Δv2-second impulse [km/s]	1.5606	0.0483
Total Δv [km/s]	1.6297	1.5990
Total time [s]	10674	12144

In terms of cost, the second maneuver point is more convenient, even if the Δt is slightly bigger. The total cost of this strategy is 9.7591 km/s and the total time used is 34105 seconds (9h:28m:25s).

There are two main criticalities in this transfer: plane change maneuver's cost weights 80.56% on the total cost, and there are four impulses to be successfully completed in order to reach the final orbit. Since the plane change maneuver's cost decreases if it is operated far from the attractor, one possible solution is to change the orbit's shape (eccentricity and semimajor axis) before the plane change maneuver. This is the idea behind the **2. optimized standard strategy.** The first maneuver is the bielliptic transfer; the two possible maneuver points are under reported:

	First maneuver point (perigee)	**Second maneuver point (apogee)**
ECI position [km]	(-382; 6696.3; 2230.2)	(496.5; -8702.9; -2898.5)
Δt (initial point→maneuver point) [s]	5501	1855
Δt (maneuver) [s]	3527	8642
Δv1-first impulse [km/s]	0.0691	1.5507
Δv2-second impulse [km/s]	1.5606	0.0483
Total Δv [km/s]	1.6297	1.5990
Total time [s]	9028	10497

We choose the apogee-apogee maneuver, as it is more efficient in terms of cost. The plane change maneuver will be less expensive now, as reported in the table below.

	First intersection point	**Second intersection point**
ECI intersection position [km]	(-3800.6; 6774.6; -8546.7)	(4251.7; -7578.8; 9561.3)
Δt (apogee→intersection point) [s]	10418	22880
Δv [km/s]	6.8507	6.1238

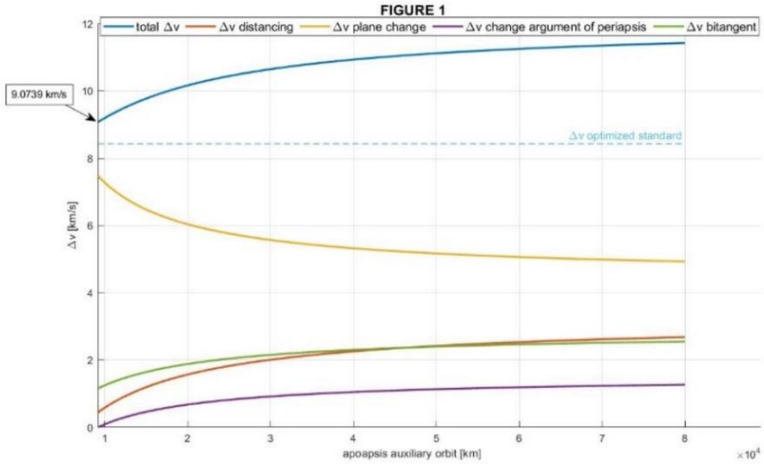
Maneuvering at the second intersection point makes the waiting time higher, but reduces the cost; for this reason, we choose to operate the maneuver at the second intersection point.
The final argument of perigee is reached with a maneuver at the second point of intersection, as it minimizes the time, and the cost does not change.

	First intersection point	**Second intersection point**
ECI intersection position [km]	(-7517; 11442; -13957)	(3407.7; -5187; 6327)
Δt (plane change point →intersection point) [s]	10148	2385
Δt (maneuver point →final point) [s]	15703	7940
Total time [s]	25851	10325
Δv [km/s]	0.7062	0.7062

We obtain a total time of 43701 seconds (12h:8m:21s), and a Δv of 8.4290 km/s.

As seen above, the most expensive maneuver is the plane change; its cost is lower as we perform it far from the Earth. For this reason, we thought that we could reduce the total cost of the transfer by moving from the apogee of the initial orbit to an auxiliary elliptic orbit with periapsis equal to the apoapsis of the first orbit; then executing the plane change on this orbit. We can now reach the final orbit by changing the argument of periapsis and then the shape of the orbit with a bitangent maneuver, as done in the standard transfer. However, as reported in FIGURE 1, increasing the auxiliary orbit's apoapsis is not convenient, in the limits of Earth's sphere of influence; in fact, the most efficient case is the circular auxiliary orbit, that involves a total cost of 9.0739 km/s, bigger than the total costs of the optimized strategy.

FIGURE 1



This result is due to the addiction of a new maneuver, the one needed to transfer from the initial orbit to the auxiliary orbit, and the fact that changing shape and orientation in this new orbit implies bigger costs.

Considering these results, in order to implement a more efficient transfer, we need to optimize not only the plane change but also the other maneuvers. This is the aim of the **3. transfer with circularization**: starting at the apocentre of the initial orbit, with a tangential impulse we pass to an auxiliary orbit with apoapsis equal to the apoapsis of the final orbit. At the apocenter we proceed to circularize the orbit with a tangential impulse and on this new orbit we perform the plane change. The main advantage of being on a circular orbit with the same radius as the apoapsis of the final orbit is that now we can reach the final shape and orientation with a single tangential maneuver. This is done by changing the velocity when the satellite in the circular orbit is at the point that coincides with the apocentre on the final orbit. Making all the calculations, and considering the time to reach the final point, we obtain a total Δv of 8.4197 km/s and a period of 42157 s (11h:42m:37s). Even if this strategy is less expensive than the optimized standard strategy, it requires a long period of time to be achieved, and we still must successfully complete four impulses. To reduce the risk of failing an impulse, we elaborated the **4. two-impulse transfer,** whose aim is to minimize the number of impulses; for this reason, we decide to operate only two maneuvers, which is the minimum number possible for a transfer between two non-intersecting orbits like the given ones. Furthermore, reducing the number of impulses reduces also the time spent on transfer orbits waiting for the next maneuver point, and thus the total time spent.

In detail we find the two points of intersection between the final orbit and the initial orbital plane; therefore, at perigee of the initial orbit, we make a tangent shape change into an elliptical transfer orbit that will bring the satellite to one of the points of intersection evaluated earlier (we will later discuss which one is more efficient). At this point we change the velocity of the satellite in order to reach the velocity needed in the final orbit at that point, changing plane, shape and orientation in a single boost.

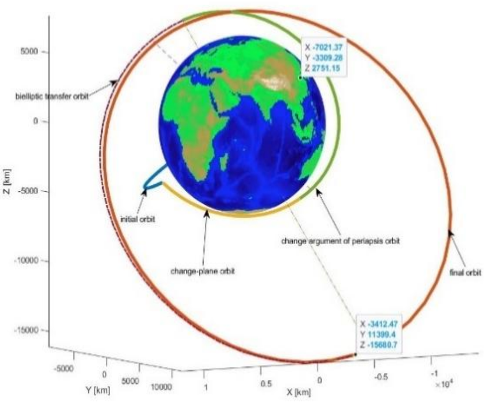
Making the calculations we find the following results:

	First intersection point	Second intersection point
ECI intersection position [km]	(11600, 634, -6008)	(-10133, -553, 5248)
t (initial point→perigee) [s]	5501	5501
Δv first maneuver [km/s]	1.7165	2.1150
t (perigee→intersection) [s]	28979	1421
Δv second maneuver [km/s]	7.5884	6.1256
t (intersection→final point) [s]	5482	10001
total Δv [km/s]	9.3049	8.2406
total time [s]	39962	16923

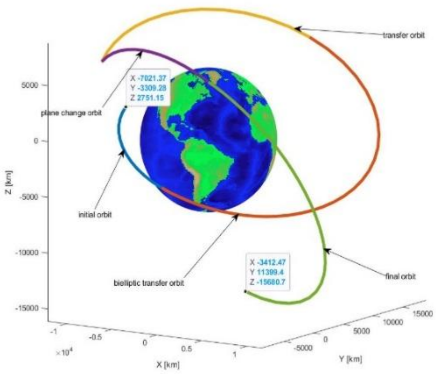
We can clearly see that using the second intersection point is convenient in both terms of time and cost.

4.3

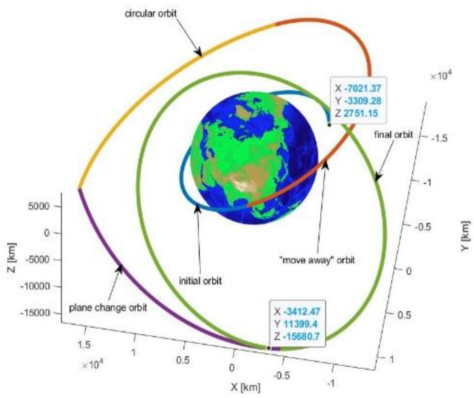
STANDARD TRANSFER



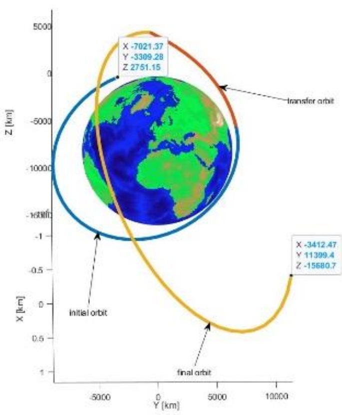
OPTIMIZED STANDARD TRANSFER



TRANSFER WITH CIRCULARIZATION

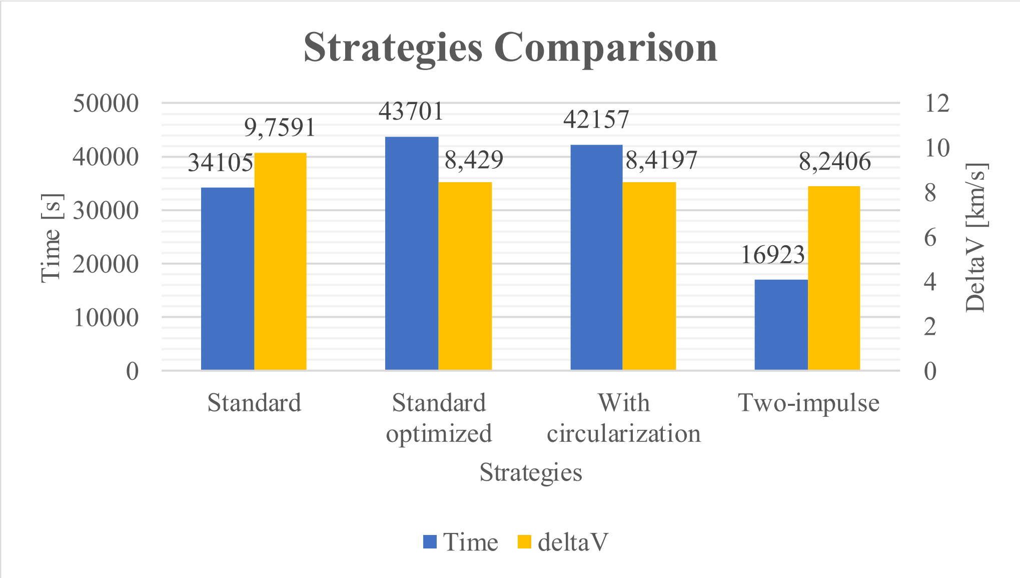


TWO-IMPULSE TRANSFER



5. Conclusions

As said in the introduction, our work's aim is to select the most efficient orbital transfer. The proposed strategies are compared in the graph below:



Considering this comparison, and the fact that the first three maneuvers require four impulses to be successfully achieved, the most efficient transfer strategy is the two-impulse transfer; it connects the two points given using the lowest amount of fuel, the minimum time and the lowest number of impulses.

6. Appendix

Transfer 1 (standard strategy)

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8127.3040	0.1303	0.5507	1.0556	0.6472	1.8021	-
3809	8127.3040	0.1303	0.5507	1.0556	0.6472	4.5649	7.8619
	8127.3040	0.1303	1.0030	2.8600	5.4361	4.5649	
5356	8127.3040	0.1303	1.0030	2.8600	5.4361	6.1204	0.2983
	8127.3040	0.1303	1.0030	2.8600	5.1106	0.1628	
8858	8127.3040	0.1303	1.0030	2.8600	5.1106	3.1415	1.5507
	14448.0305	0.3642	1.0030	2.8600	1.9690	0	
17500	14448.0305	0.3642	1.0030	2.8600	1.9690	3.1416	0.0483
	14270.0000	0.3812	1.0030	2.8600	1.9690	3.1416	
34105	14270.0000	0.3812	1.0030	2.8600	1.9690	3.0770	-

Transfer 2 (optimized standard strategy)

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8127.3040	0.1303	0.5507	1.0556	0.6472	1.8021	-
1855	8127.3040	0.1303	0.5507	1.0556	0.6472	3.1416	1.5507
	14448.0305	0.3642	0.5507	1.0556	3.7888	0	
10496	14448.0305	0.3642	0.5507	1.0556	3.7888	3.1416	0.0483
	14270.0000	0.3812	0.5507	1.0556	3.7888	3.1416	
33376	14270.0000	0.3812	0.5507	1.0556	3.7888	4.5649	6.1238
	14270.0000	0.3812	1.0030	2.8600	2.2945	4.5649	
35761	14270.0000	0.3812	1.0030	2.8600	2.2945	6.1204	0.7062
	14270.0000	0.3812	1.0030	2.8600	1.9690	0.1628	
43701	14270.0000	0.3812	1.0030	2.8600	1.9690	3.0770	-

Transfer 3 (transfer with circularization)

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8127.3041	0.1303	0.5507	1.0556	0.6472	1.8021	-
5501	8127.3041	0.1303	0.5507	1.0556	0.6472	6.2832	1.1275
	13388.9976	0.4721	0.5507	1.0556	0.6472	0	
13210	13388.9976	0.4721	0.5507	1.0556	0.6472	3.1416	1.2296
	13388.9976	0	0.5507	1.0556	0	3.7888	
19448	13388.9976	0	0.5507	1.0556	0	5.2120	5.1031
	13388.9976	0	1.0030	2.8600	0	3.7178	
25552	13388.9976	0	1.0030	2.8600	0	5.1106	0.9595
	14270.0000	0.3812	1.0030	2.8600	1.9690	3.1416	
42157	14270.0000	0.3812	1.0030	2.8600	1.9690	3.0770	-

Transfer 4 (two-impulse transfer)

t (s)	a (km)	e (-)	i (rad)	Ω (rad)	ω (rad)	θ (rad)	Δv (km/s)
0	8127.3041	0.1303	0.5507	1.0556	0.6472	1.8021	-
5501	8127.3041	0.1303	0.5507	1.0556	0.6472	6.2832	2.1150
	36908.4675	0.8085	0.5507	1.0556	0.6472	0	
6922	36908.4675	0.8085	0.5507	1.0556	0.6472	1.4233	6.1256
	14270.0000	0.3812	1.0030	2.8600	1.9690	4.8904	
16923	14270.0000	0.3812	1.0030	2.8600	1.9690	3.0770	-